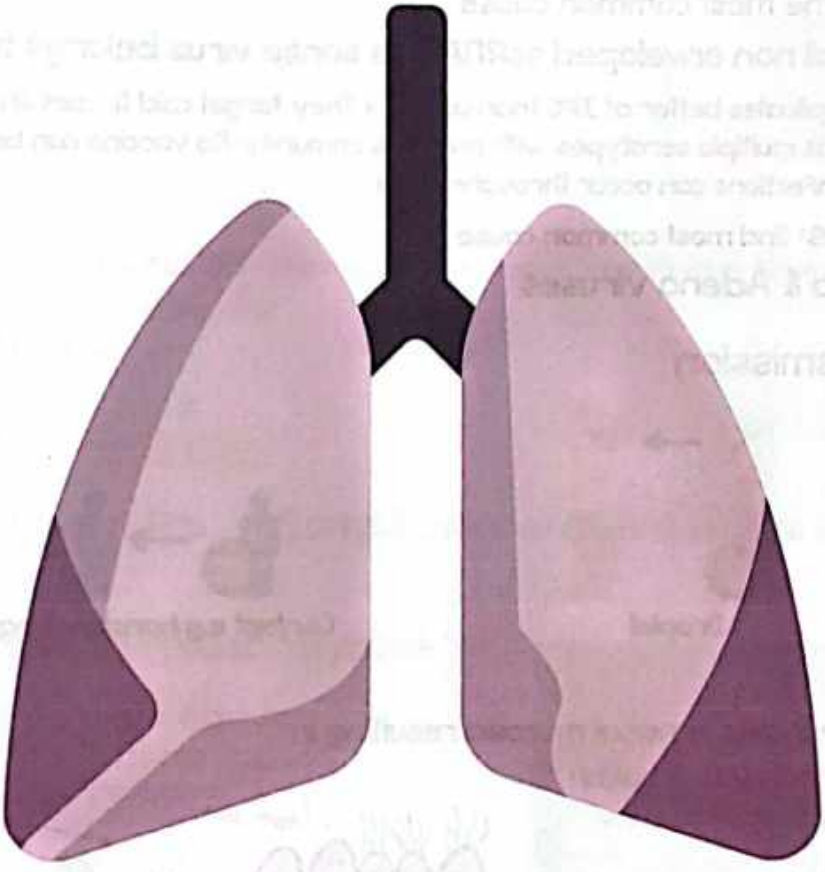
2026



MICROTUTORIALS®

RESPIRATORY MODULE

1

Scan to join Telegram group

Pulse.

DR. MUHAMMED ASHRAF

Chapter I: URT infections

I- Common Cold: a viral infection of the URT mucosa

Etiology:

I- Rhinoviruses: the most common cause

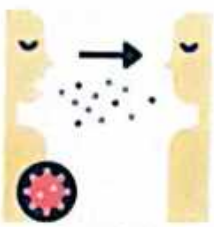
Structure: small non enveloped ssRNA +ve sense virus belongs to Picorna viruse

Features:
- it replicates better at 33°C than at 37°C = They target cold tissues like URT and conjunctiva
- it has multiple serotypes with no cross immunity: No vaccine can be developed & reinfections can occur throughout life

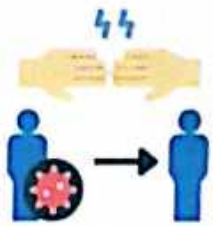
2- Coronaviruses: 2nd most common cause

3- parainfluenza & Adeno viruses

Mode of transmission



Droplet

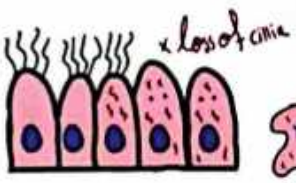


Contact e.g handshaking

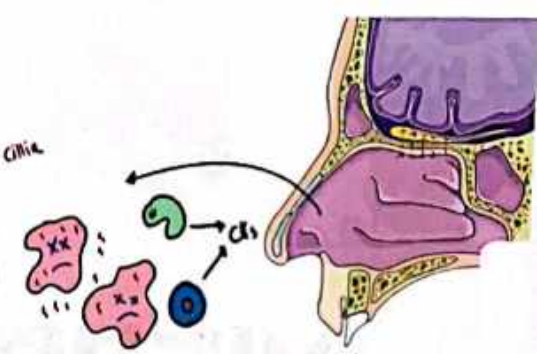
Pathogenesis

Viral particles infect cells in nasal mucosa resulting in :

I- Loss of ciliary processes on apical surfecre = 2ry bacterial infections



2- Sloughing off and death of infectd cells results in acute inflammatory reaction: released mediators & Cks increase mucus production, resulting in sneezing & congestion of nasal mucosa



Clinical Picture

- FAHM "Acute inflammation"
- Nasal congestion and rhinorrhea (Watery nasal discharge)
- Anosmia (Decreased up to loss of smelling)

Diagnosis: Clinically **Prevention:** No available vaccine

Complications: I- Local spread (Siusitis, AOM) 3- 2ry bacterial infections
2- Triggiring asthma

..ute Sinusitis: acute inflammation of paranasal sinuses

..ology:



Bacterial:

1- Strept. pneumoniae : the most common cause
2- H. influenzae
3- Moraxella catarrhalis.

Fungal: In immunocompermized

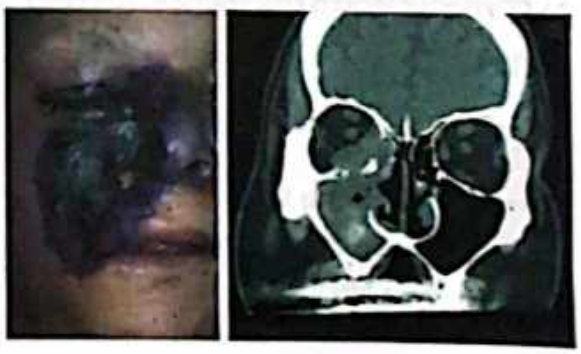
1- Aspargillus
2- Mucor

Pathogenesis

- 2ry bacterial infection of Viral URT "most common cause" resulting in Acute inflammation of paranasal sinuses characterized by:
  - > Purulent nasal discharge
  - > Nasal congestion
  - > Sinus & facial pain
  - > Headache

Diagnosis: Clinically & CT- PNS **Complications:** Chronic sinusitis & polyps formation

III- Fungal Sinusitis "Mucormycosis" :



Etiology: Mucor, Rhizopus & Absidia
Characteristics: non sepetate filamintous fungi with broad branching hyphae

Risk factors: Low immunity

- Old diabetics "Acidosis"
- iron overload
- Leukemia & lymphoma patients
- Corticosteriods treatment
- Severe burns

Pathogenesis & C/P

1- inhaled fungal spores germinate inside nasal cavitey and PNS

2- Fungal hyphae invade Blood vessels causing thrombosis e.g CST and Artery thrombosis Leading to ischemic necrosis and tissue destruction



3- C/P include: Bloody nasal discharge, affected facial area appears red and sowllen "Orbital cellulitis" Then turns into black due to ischemic necrosis and tissue destruction

4- Disease progresses rapidly invading cranial tisssues and brain



Lab diagnosis of mucormycosis:

- Direct microscopic examination of Nasal discharge or biopsy: Reveals non sepetate filamintou with broad branching hyphae

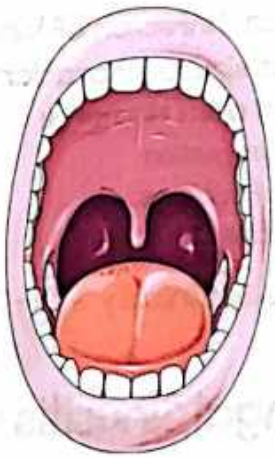
Treatment of mucormycosis

- Correct predisposing factors: Control diabetes , adjust immunosuppressive drugs dose
- Surgical debridment
- Systemic atifunglas: Amphotercin B

III. Pharyngitis and Tonsillitis

Tonsillitis: inflammation of the palatine tonsils
Pharyngitis: inflammation of the pharynx, which is in the back of the throat.

Etiology:



A- Viral "70% of cases"

- Rhinoviruses and Coronaviruses.
- Adenoviruses.
- Influenza and parainfluenza viruses.
- Coxsackie A virus.
- Epstein-Barr virus & Cytomegalovirus.

B- Bacterial

- Strept pyogenes "most important bacterial cause"
- Corynebacterium diphtheriae, Borrelia vincentii & Fusobacteria

C/P



- Sore throat that worsens during swallowing.
- FAHM



- Enlarged draining LNs

I- Coxsackie virus:

Structure: small non-enveloped ssRNA +ve sense virus. It belongs to picornaviruses

C/P:

Hand, foot and Mouth Disease:

- Fever and vesicular lesions on palate, hand and feet
- caused by Coxsackie A6 strain

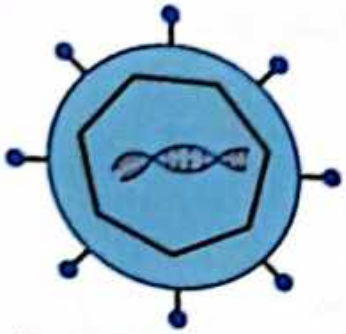


Herpangina
sudden onset of Fever and vesiculo-papullar lesions on tonsils and palate



...stein Barr Virus "EBV"

...tructure:



Structure: DsDNA Enveloped virus
Family: Herpesviruses

MOT: Saliva

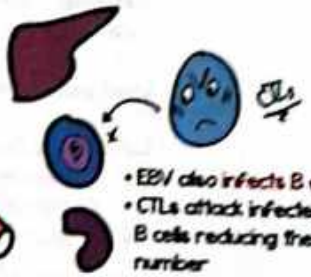
- EBV replicates in pharyngeal epithelial cells and adjacent LNs causing pharyngitis and sore throat



Pathogenesis:



- Viremia occur • Fever
- viral particles reach RES cells "liver and spleen" Causing Hepatosplenomegaly



- EBV also infects B cells
- CTLs attack infected B cells reducing their number

C/P

Symptoms of Infectious mononucleosis

IP: 1-2 months ?!

- Sore throat
- Fever & Rash
- Hepatospleenomegaly
- +- Jaundice

Complications: Latent infection

- Integration of Viral DNA into host chromosome in:

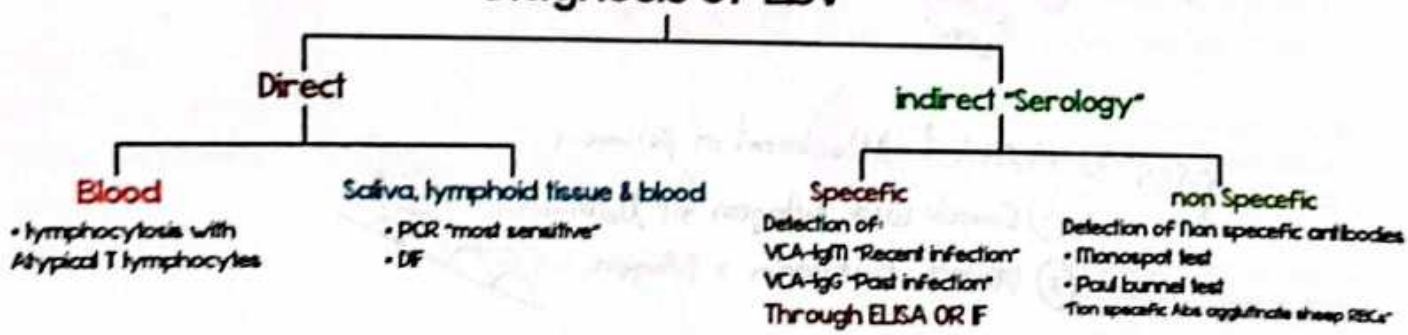


B lymphocytes:
Burkitts Lymphoma



Pharyngeal epithelial cells:
nasopharyngeal carcinoma

Diagnosis of EBV



3- Cytomegalo Virus "CMV"

- Double stranded DNA enveloped virus belongs to Herpesviridae
- Causes infectious mononucleosis like syndrome with -ve Non-Specific tests
- The virus remains latent in mononuclear cells to be reactivated with immunosuppression leading to life threatening pneumonia.

Diagnosis:

Specimens: Broncho Alveolar lavage or mononuclear cells from blood
Direct Methods: Detection of viral Ag via DIF or ELISA
Indirect methods (Serology): Detection of Anti CMV-IgM

III. Acute Otitis Media

An acute infection of the middle ear, caused mainly by:

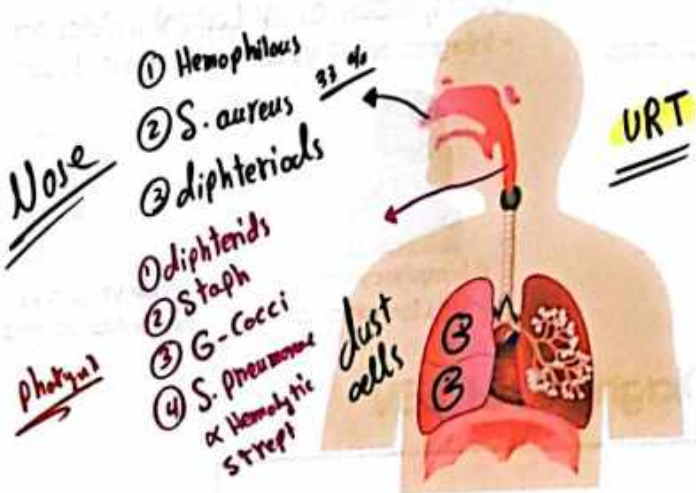
Bacterial pathogens: *St. pneumoniae (most common), H. influenzae, Moraxella catarrhalis.*

Viral pathogens: Respiratory syncytial virus, Coronaviruses, Rhinoviruses.

Laboratory Diagnosis:

- Mainly clinical.
- Culture of discharge.
- Needle aspiration of middle ear fluid.

Respiratory Flora



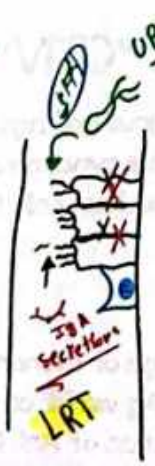
pros ⊕ ① prevent Attachment of pathogens
② Compete with pathogens for Nutrition
③ produce Bactiriocins x Pathogens



Cons ⊖ opportunistic infections

Why LRT is Sterile?

① Ciliary Motion
② Mucus + IgA
③ MQ





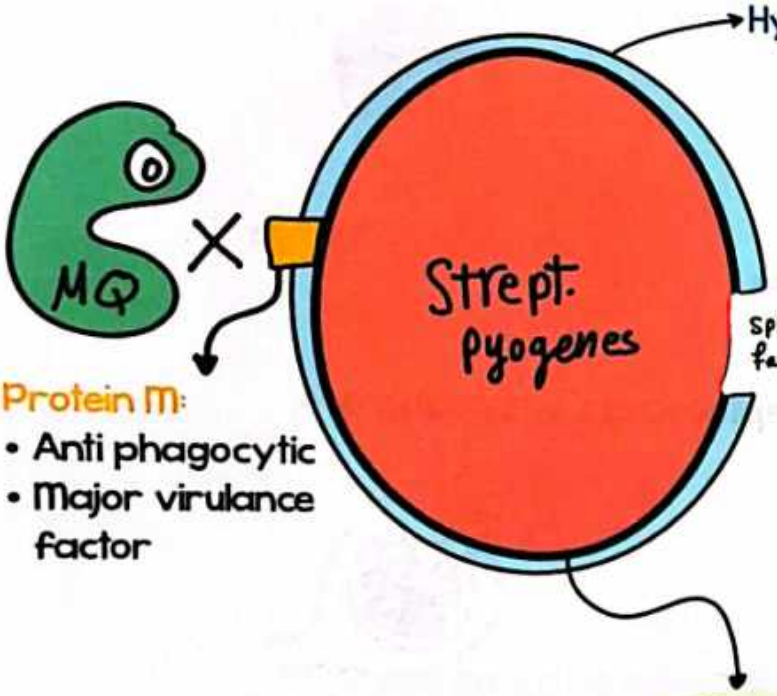
...eptococcal Pharyngitis

...rept pyogenes: Gram +ve cocci arranged in chains, Catalse -ve, non motile, non spore forming , Beta hemolytic on blood agar "Group A beta hemolytic strept."

Strept. pyogenes Diseases

- **Pyogenic diseases** — results from Local inflammation
- **Toxin mediated** — Exotoxin production = sytemic symptoms
- **immune mediated** — Due to Antibody cross reaction

II- Strept. pyogenes Virulance factors



Deoxyribonuclease "DNAse" : Breaks down DNA in exudate & liquify pus

hyaluronidase: Breaks down ground substance in connective tissue

Streptokinase "Fibrinolysin:
- Transforms plasminogen into plasmin which breaks down fibrin and clots in exudate
- It can be used clinically in cases of pulmonary embolism and thrombosis

Toxins

Pyrogenic Exotoxins (Erythrogenic toxins)
- They are 3 Antigenically distinct Exotoxins
- they can Act as SuperAntigens & cause strept. toxic shock syndrome
- Type A is coded by lysogenic phage
- Types A & C cause scarlet fever (Erythrogenic toxins)

Streptolysins (Hemolysins)
- Lyse membranes of erythrocytes

Clinical picture of Streptococcal pharyngitis:

I- Pyogenic inflammation:

General: FAHM "High grade fever", Enlarged Cervical LNs + Tachycardia and Leukocytosis

Local:



• Acute follicular Tonsilitis
Pus and debries in crypts



• Peritonsellar abcess "Quenzy"
With uvula deviated to the healthy side

II-Toxin mediated: Erythrogenic toxins cause scarlet fever:



Skin Rash:

- Appears on trunk then extremities after 24 hours of illness
- Disappear after a week with skin sloughing



Strawberry tounge

II-Immune mediated "Sterile or Post streptococcal Complications" :



Rhumatic fever



Post Streptococcla GN

Diagnosis:

Spicemen: Throat swab

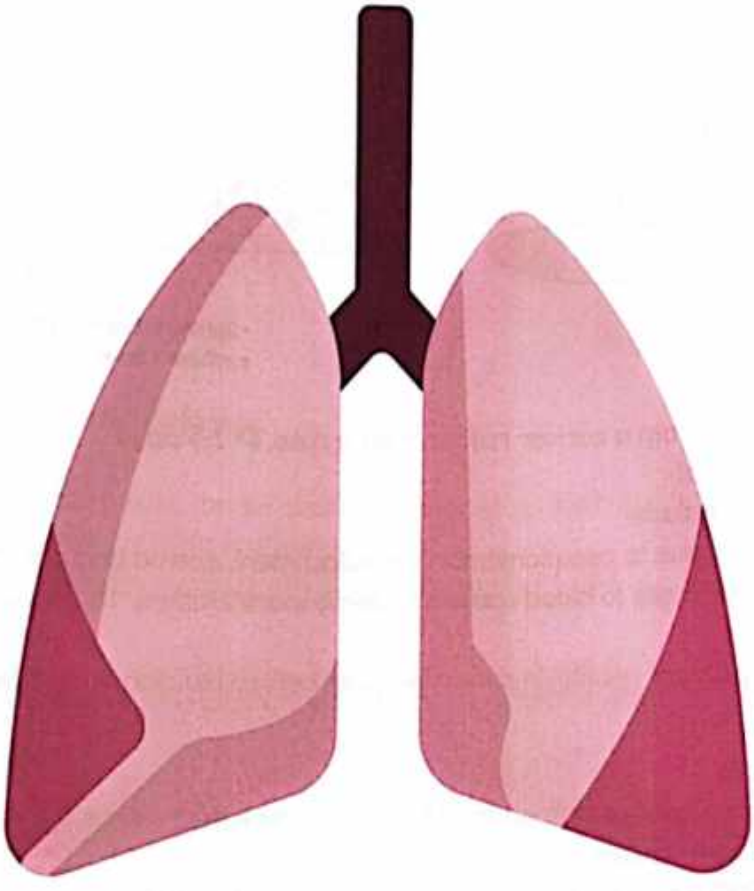
Direct microscopy: Gram +ve cocci in chains

Isolation:

- Catalase -ve
- Beta hemolytic on Blood agar
- Bacitracin sensitive

Detection of Group A Ag: using ELISA or Latex agglutination

2025



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RESPIRATORY MODULE

2

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DR. MUHAMMED ASHRAF

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Chapter 2: URT infections 2

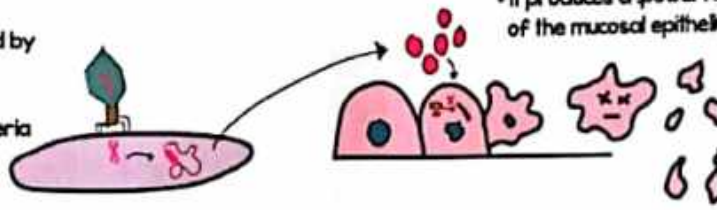
I- Diphtheria

Etiology:

Corynbacterium diphtheria: Gram +ve bacilli, aerobic, non spre forming, non motile

Pathogenesis

C. diphtheria can by lysogenized by bacteriophage "Prophage is integrated into bacterial chromosome to produce Diphtheria Exotoxin "Lysogenic Conversion"



- C. diphtheriae implants on the mucosa of the URT
- It produces a powerful exotoxin that causes necrosis of the mucosal epithelium leading to formation of dirty grey pseudo-membrane with abundant fibrinopurulent exudate.
- No mucosal penetration = no bacteremia

- Diphtheria Exotoxin: inhibits EF-2 = inhibits Protein synthesis = tissue necrosis

Clinical Picture

- Most infections come from a carrier rather than a case, IP: 2-5 days
- S/S: FAHM + sore throat
- Major complications include:
  - > Airway obstruction: due to pseudomembrane detachment, edema and periphral nuritis
  - > Toxemia: Exotoxin diffuses to blood causing systemic manifistations "Myocarditis, peripharal nuritis"

Diagnosis: Mainly Clinical, Antitoxin should be given before lab diagnosis "Life saving"

Lab diagnosis:
Spicemen: Throat swab
Direct microscopy: Gram positive club-shaped bacilli "Chinese letter"
Isolation: on Loeffler's serum medium or Blood tellurite
Detection of Diphtheria toxin: using ELISA, Elek's test or pcr for Tox gene

Prevention:

- Diphtheria toxoid is given in combination with pertussis vaccine and tetanus toxoid (DPT)

Notes:

Vincent's angina "Trench mouth"

Etiology:

- **Borrelia vincentil:** Loosly coiled irregular spirochetes
- **Fusobacteria:** Gram negative anaerobic short fusiform or filamentous bacilli

Pathogenesis

- They are usually commensals in mouth
- Decreased immunity due to: malnutrition, mucosal damage or infections "HSV" cause them to increase in number

Clinical Picture

- Ulcerative gingivostomatitis (trench mouth): ulcers on the tonsils and the gum with pseudomembrane containing pus cells



Diagnosis:

Spicemen: swab from pseudomembrane.

Direct microscopy:

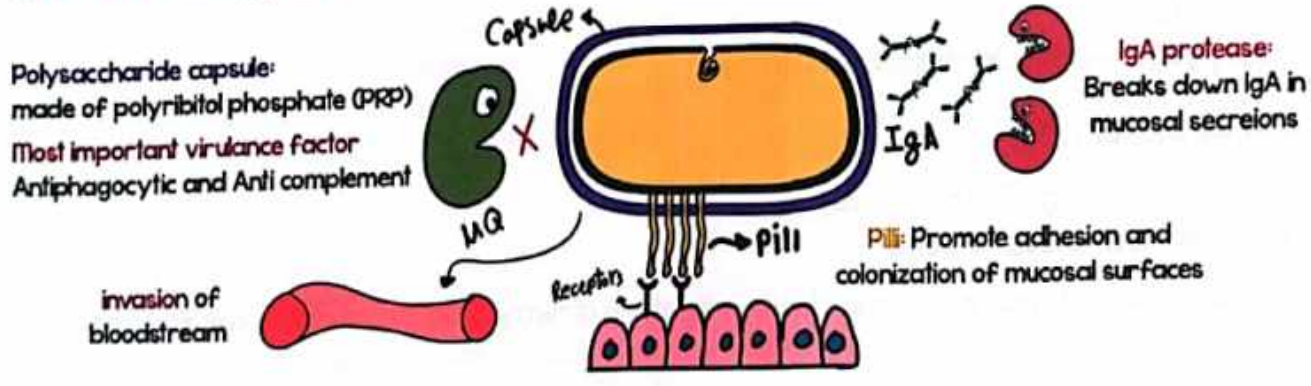
- B. vencintii: Gram -ve helical rods, can be also stained by gimsa stain
- Fusiforms: Gram -ve short fusiform or filamentous bacilli

Isolation: in broth enriched with ascitic fluid & incubated anaerobically

III- Epiglottitis: Life-threatening condition that occurs when the epiglottis swell and block airway

Etiology:

H. influenzae type b: most common cause, it's virulance factors include:



H. influenza: Small gram -ve coccobacilli "short rods"

- Most virulant strains have PS capusle, other weak strains are non-capsulated
- there are 6 serotypes of Capsule Ag (A-F), type b is the most virulant
- isolation of the organism on laboratory media requires the addition of two components: hemin (factor X) and NAD (factor V) = Chocolate agar

Mode of transmission
- Droplet or direct contact

Clinical Picture
- FAHM, Severe sore throat
- Odynophagia & Dysphagia "Painfull difficult swallowing"
- Stridor (Difficult breathing and abnormal breathing sound)
- Airway obstruction "Fatal"

Diagnosis: Clinically + Confirmed by isolation on culture media:

Sample: Sputum

Direct microscopic examination: G-ve pleomorphic bacilli

Isolation:
- Type of media: Chocolate Agar with 5-10% CO2 & Factor V & X
- Satellatism: H. influenza grow on blood agar around colonies of S. aureus as S. aureus produces factor V and factor X required by H. influenza

Other direct methods: PCR or Capsule Ag detection by ELISA

Prevention: Conjugated Polysaccharide vaccine contains capsule of Type b (Hi-b) given between 2~18 months of age

IV- Laryngitis & Tracheitis

Etiology:

A- Viral
- Parainfluenza viruses "Most common cause"
- Adenoviruses, influenza viruses & corona viruses

B- Bacterial
- B. pertussis
- Strept pyogenes, M. catarthalis, H. influenzae

Clinical picture:

Tracheitis: Dry cough, Retrosternal pain

Laryngitis: Hoarsness of voice

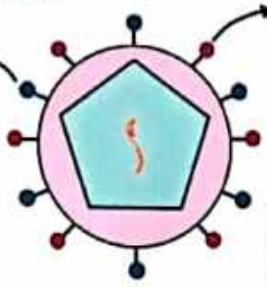
- Both can cause stridorin children due to immature airway (Non expandable cartilage rings)

Notes:

Para influenza viruses "HPIV" :

Structure: Spherical, Enveloped, ssRNA "Unsegmented & -ve sense"
Family: Paramyxoviruses
Envelope has Two important Glycoproteins:

Fusion (F) protein:
- Promotes fusion of Viral envelope with the Host's cell membrane which is an initial step in infection
- Also causes fusion between infected cells and syncytium formation



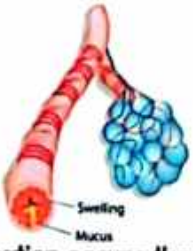
Hemagglutinin-Neuraminidase (HN):
- Helps in viral attachment to host cells
- also causes hemadsorption and hemagglutination when virus is isolated on tissue culture

HPIV has 4 serotypes, HPIV-1, HPIV-2 cause Croup in children (Acute laryngiotracheobronchitis)

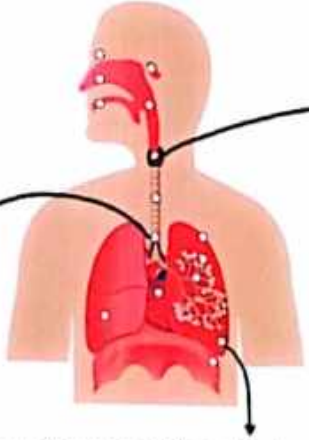
Clinical Picture

Adults: Common cold (URTI symptoms)
Children: Croup "Acute laryngotracheobronchitis", Bronchiolitis & pneumonia (LRTI Symptoms)

Pathogenesis



Virus multiplication occurs throughout the tracheobronchial tree, inducing production of mucus.





vocal cords of the larynx become grossly swollen, obstructing air inflow, which is manifested by inspiratory stridor.

In infants and young children, the bronchi, bronchioles and lungs are occasionally involved due to:
- Small size & Immaturity of the airways
- immature immunity



Diagnosis

- Mainly clinical
- Lab diagnosis:

Specimens: Nasal washes, nasopharyngeal aspirate and bronchoalveolar Lavage (BAL).
Detection and demonstration of Viral Ag: by DIF
Viral Isolation to demonstrate:
- CPE: syncytia formation.
- hemadsorption

Serology: ELISA for detection of IgM or IgG